ITI0402 Loogika, sügis 2026
Harjutus 3
Ülesanded

1. • $\neg(q \supset \neg p) \supset r \vee \neg q$
Semantiline tabel:

$\cancel{\neg(q \supset \neg p) \supset r \vee \neg q}$
$\neg(q \supset \neg p)$
$\cancel{r \vee \neg q}$
$\cancel{q \supset \neg p}$
q
$\cancel{\neg p}$
$\cancel{r}$
$\cancel{\neg q}$
p
q
✓

Väärav väärtustus: $\begin{cases} p \mapsto \mathbf{t} \\ q \mapsto \mathbf{t} \\ r \mapsto \mathbf{f} \end{cases}$.

KNK: $\neg p \vee \neg q \vee r$.

• $(p \vee \neg q) \supset \neg(q \wedge \neg p)$
Semantiline tabel:

$\cancel{(p \vee \neg q) \supset \neg(q \wedge \neg p)}$
$p \vee \neg q$
$\cancel{\neg(q \wedge \neg p)}$
$q \wedge \neg p$
q
$\neg p$
$\cancel{p}$

p	$\neg q$
×	$\cancel{q}$
	×

Tautoloogia.

• $((p \supset q) \supset p) \supset p$
Semantiline tabel:

$\cancel{((p \supset q) \supset p) \supset p}$
$(p \supset q) \supset p$
$\cancel{p}$

$\cancel{p \supset q}$	p
p	×
$\cancel{q}$	
×	

Tautoloogia.

2. • $(p \supset q) \wedge (\neg q \supset \neg p)$

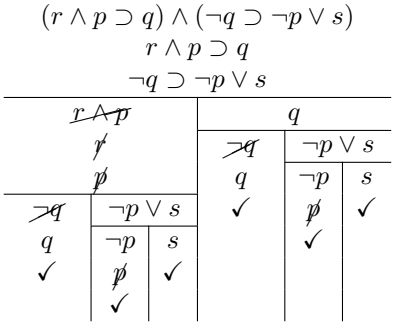
$(p \supset q) \wedge (\neg q \supset \neg p)$
$p \supset q$
$\neg q \supset \neg p$

$\cancel{p}$		q	
$\cancel{\neg q}$	$\neg p$	$\cancel{\neg q}$	$\neg p$
q	$\cancel{p}$	q	$\cancel{p}$
✓	✓	✓	✓

Kehtestavad väärtustused: $\begin{cases} p \mapsto \mathbf{f} \\ q \mapsto \mathbf{t} \end{cases}$, $\{\ p \mapsto \mathbf{f}$, $\{\ q \mapsto \mathbf{t}$, $\begin{cases} p \mapsto \mathbf{f} \\ q \mapsto \mathbf{t} \end{cases}$
(kus 1. ja 4. on samad ja nii 2. kui ka 3. katavad nad).

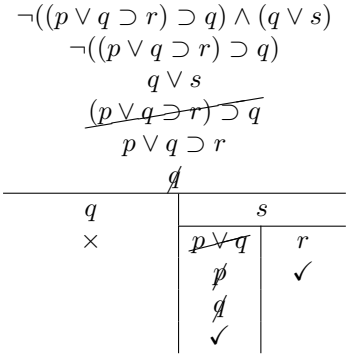
DNK: $(\neg p \wedge q) \vee \neg p \vee q \vee (\neg p \wedge q)$.
Väiksem DNK: $\neg p \vee q$.

- $(r \wedge p \supset q) \wedge (\neg q \supset \neg p \vee s)$



Kehtestavad väärtustused: $\left\{ \begin{array}{l} p \mapsto \mathbf{f} \\ q \mapsto \mathbf{t} \\ r \mapsto \mathbf{f} \end{array} \right.$, $\left\{ \begin{array}{l} p \mapsto \mathbf{f} \\ r \mapsto \mathbf{f} \end{array} \right.$, $\left\{ \begin{array}{l} p \mapsto \mathbf{f} \\ r \mapsto \mathbf{f} \\ s \mapsto \mathbf{t} \end{array} \right.$, $\{ \; q \mapsto \mathbf{t}$, $\left\{ \begin{array}{l} p \mapsto \mathbf{f} \\ q \mapsto \mathbf{t} \end{array} \right.$, $\left\{ \begin{array}{l} q \mapsto \mathbf{t} \\ s \mapsto \mathbf{t} \end{array} \right.$
(siin 2. katab 1. ja 3. ning 4. katab 5. ja 6.).
DNK: $(\neg p \wedge q \wedge \neg r) \vee (\neg p \wedge \neg r) \vee (\neg p \wedge \neg r \wedge s) \vee q \vee (\neg p \wedge q) \vee (q \vee s)$.
Väiksem DNK: $(\neg p \wedge \neg r) \vee q$.

- $\neg((p \vee q \supset r) \supset q) \wedge (q \vee s)$



Kehtestavad väärtustused: $\left\{ \begin{array}{l} p \mapsto \mathbf{f} \\ q \mapsto \mathbf{f} \\ s \mapsto \mathbf{t} \end{array} \right.$, $\left\{ \begin{array}{l} r \mapsto \mathbf{t} \\ s \mapsto \mathbf{t} \end{array} \right.$.
DNK: $(\neg p \wedge \neg q \wedge s) \vee (r \wedge s)$.

3.

